
Old Exam Questions

AI in Life Sciences

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How this file is organized

Questions come from two sittings: the **2024** exam (25 June 2024, Sections 1–6) and the **2026** exam (25 June 2026, Sections 7–10). The 2024 items are *multiple-select* (“which of the following is/are true” means **one or more** of the four options are correct). The 2026 items are *single-best-answer* (exactly **one** of the three options is correct). To avoid reproducing the originals verbatim, the question wording has been paraphrased while the answer options are kept intact. Questions are grouped by topic rather than in their original random order. The full answer key is at the end — the question boxes themselves are intentionally left blank so the file doubles as a practice set.

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Drug Discovery, Molecules & Representations

Question 1: SMILES strings

Regarding the SMILES notation for molecules, which of the following statements hold?

- ☐ SMILES stands for *Simplified Molecular Input Line Entry System*.
- ☐ SMILES strings represent molecules in a linear sequence of characters.
- ☐ SMILES cannot be converted into a molecular graph.
- ☐ SMILES is always unique for each molecule, ensuring no two different molecules share the same SMILES.

Question 2: Reading a SMILES string

Consider the molecule *pyruvic acid* (2-oxopropanoic acid), i.e. a methyl group followed by a ketone carbonyl and then a carboxylic acid ($\text{CH}_3\text{-C(=O)-C(=O)-OH}$). Which of the SMILES strings below correctly encode this molecule?

- ☐ O=C(C(=O)O)C
- ☐ CC(=O)C(=O)O
- ☐ CCC000
- ☐ OCCOO

Question 3: Molecular fingerprints

Which of the following statements about molecular fingerprints are accurate?

- ☐ A unique molecular graph can be constructed from them.
- ☐ They are binary or numerical representations of molecular structures.
- ☐ They are used for similarity searching and machine learning tasks.
- ☐ They provide a 3D representation of molecules.

Question 4: Bioassays

Select the correct statements about bioassays.

- ☐ They can be used to measure the biological response or function elicited by a substance.
- ☐ They typically test the substance at various concentrations.
- ☐ They are wet-lab procedures to acquire knowledge of a compound in a biological context.
- ☐ They are typically tested *in vitro*, i.e. in a living organism.

Question 5: ADME properties

Which statements about ADME (Absorption, Distribution, Metabolism, and Excretion) properties are correct?

- ☐ Lipinski's rule of five can be used as a proxy for ADME properties.
- ☐ Good ADME properties guarantee that a compound will be an effective drug.

- ☐ ADME properties such as clearance are often difficult to model.
- ☐ ADME properties are critical in determining the pharmacokinetics of a drug.

Question 6: Characteristics of QSAR / bioactivity data

Which of the following correctly characterise typical QSAR / bioactivity datasets?

- ☐ QSAR datasets usually have data points in the order of thousands or millions.
- ☐ QSAR datasets typically have a balanced distribution of active and inactive compounds.
- ☐ QSAR datasets often contain series of structurally related compounds.
- ☐ QSAR data often includes noise due to biological variability and experimental inconsistencies.

Question 7: Applications of QSAR modeling

For which of the following applications is QSAR modeling relevant?

- ☐ Creating 3D protein structure models.
- ☐ Identifying potential drug candidates.
- ☐ Predicting the toxicity of new chemical compounds.
- ☐ Mapping the genome sequences of organisms.

Virtual Screening & QSAR Modeling

Question 8: Ligand-based virtual screening

Which statements about ligand-based virtual screening are correct?

- ☐ Molecular fingerprints are commonly used in ligand-based virtual screening.
- ☐ Ligand-based virtual screening is not suitable for predicting the bioactivity of unknown compounds.
- ☐ Ligand-based virtual screening requires a 3D structure of the target protein.
- ☐ Ligand-based virtual screening typically only considers bioactivity predictions to select compounds.

Question 9: Data splits in QSAR model evaluation

When evaluating QSAR models, which of the following statements about data-splitting strategies hold?

- ☐ Temporal splits help evaluate the model's performance on future, unseen data by simulating real-world data acquisition over time.
- ☐ Generally, a random split is the best method for evaluating QSAR models as it eliminates biases.
- ☐ Cross-validation is rarely used in the evaluation of QSAR due to computational cost.

- ☐ Cluster splits group similar compounds together, ensuring that structurally similar compounds are not both in the training and test sets.

Question 10: Early-recognition metrics in virtual screening

Because only the top of a virtual-screening ranking is followed up experimentally, recognising actives early matters most. Which of the following metrics are specifically designed to reward strong early recognition?

- ☐ Precision-Recall Curve.
- ☐ BEDROC (Boltzmann-Enhanced Discrimination of ROC).
- ☐ Average Precision (AP).
- ☐ Enrichment Factor (EF).

Question 11: Target identification / deconvolution

Which statements about target identification / deconvolution are correct?

- ☐ For a given molecule, it consists of identifying to which protein it will most likely bind if multiple are present simultaneously.
- ☐ Multi-class classification can be used for it.
- ☐ Multi-task classification can be used for it.
- ☐ For a given molecule, it consists of identifying all the proteins it will bind if multiple are present simultaneously.

Question 12: Drug synergy and antagonism

Which of the following statements about drug synergy and antagonism are correct?

- ☐ Drug synergy occurs when the combined effect of two or more drugs is greater than the sum of their individual effects.
- ☐ The Loewe additivity model is commonly used to assess drug synergy.
- ☐ Antagonistic effects occur when the combined effect of two or more drugs is greater than the sum of their individual effects.
- ☐ Drug synergy / antagonism evaluation is only applicable to chemically similar drugs.

Question 13: Random Forests vs. Graph Neural Networks

Comparing Random Forests (RF) and Graph Neural Networks (GNNs) for molecular modeling, which statements are correct?

- ☐ GNNs are inherently better for QSAR modeling because they naturally capture the graph-like structure of molecules, leading to more accurate predictions in all cases.
- ☐ RF models are less prone to overfitting compared to GNNs, making them a more reliable choice for datasets with a large number of features but limited samples.
- ☐ RF models are typically more interpretable than GNNs, making it easier to understand the contribution of individual molecular features to the model's predictions.

- ☐ GNNs are less sensitive to the quality and variability of the input data compared to RFs, making them more consistent in performance across different datasets.

Graph Neural Networks & Advanced Methods

Question 14: Molecular graphs and graph neural networks

Which statements about molecular graphs and graph neural networks are correct?

- ☐ A molecular graph can be represented by two matrices: one storing the node features and another storing the edges between the nodes.
- ☐ Node features can include atomic number, hybridization state, and formal charge.
- ☐ In a GNN only the node features are used to make the final predictions.
- ☐ The information about the bond type (e.g. single bond, double bond) cannot be included in a GNN.

Question 15: Limitations of GNNs in molecular modeling

Which of the following are genuine limitations of GNNs in the context of molecular modeling?

- ☐ GNNs are incapable of predicting any molecular properties accurately.
- ☐ GNNs can have difficulty capturing long-range interactions between atoms.
- ☐ GNNs may struggle with very large and complex molecular graphs.
- ☐ GNNs require a significant amount of computational resources for training.

Question 16: Multi-modal contrastive learning

Which statements about multi-modal contrastive learning are correct?

- ☐ Multi-modal contrastive learning encodes multiple modalities (e.g. molecules and proteins) into a shared latent space.
- ☐ Multi-modal contrastive learning is a self-supervised learning technique.
- ☐ Multi-modal contrastive learning concentrates (latent) representations from multiple modalities (e.g. molecules and proteins) to make predictions.
- ☐ Multi-modal contrastive learning is an unsupervised learning technique.

Question 17: Multiple instance learning (MIL)

Which of the following statements about multiple instance learning (MIL) hold?

- ☐ MIL involves training on sets or bags of instances where each instance is individually labelled.
- ☐ MIL requires the set/bag size to be fixed.
- ☐ MIL involves training on sets or bags of instances that have a single label.
- ☐ MIL models should be invariant to the permutation of instances.

Generative Models & Chemical Synthesis

Question 18: Generative models for small molecules

Which statements about generative models for small molecules are correct?

- ☐ As the syntax of SMILES strings is complicated and with long dependencies, generators of SMILES representations will produce less than 10% valid SMILES strings.
- ☐ Generative models are used to explore the known chemical space of synthesized molecules.
- ☐ Generative adversarial networks cannot be used to generate small molecules.
- ☐ Autoregressive models can be used to generate SMILES representations of small molecules.

Question 19: Goal-directed molecule generation

Which of the following correctly describe goal-directed molecule generation?

- ☐ Goal-directed generation focuses only on replicating existing molecular structures with desired properties.
- ☐ Goal-directed generation aims to create molecules with similar properties to the training set.
- ☐ Reinforcement learning can be used to optimize goal-directed molecule generation.
- ☐ Goal-directed generation aims to create molecules with specific desired properties.

Question 20: Fréchet ChemNet Distance (FCD)

Which statements about the Fréchet ChemNet Distance (FCD) and how it is computed are correct?

- ☐ FCD measures the distance between individual atoms of molecules.
- ☐ FCD is calculated by comparing the means and covariances of the latent representations of molecules.
- ☐ FCD is calculated directly on the raw molecular structures without any transformations.
- ☐ FCD is a metric used to compare the distribution of generated molecules to that of real molecules.

Question 21: AI for chemical reactions

Which of the following statements about machine learning for chemical reactions are correct?

- ☐ Some methods rely on so-called reaction templates, which are transformation rules for molecules.
- ☐ Deep learning methods that predict outcomes of chemical reactions can be trained on databases containing known chemical reactions.
- ☐ Transformer architectures have been used to predict outcomes of chemical reactions.
- ☐ Predicting chemical reactions with machine learning requires data from quantum mechanical calculations due to bond changes and electron movements.

Question 22: Synthesis planning

Which statements about synthesis planning are correct?

- ☐ Synthesis planning involves designing a sequence of chemical reactions to produce a target compound.
- ☐ Retrosynthetic analysis is a key technique used in synthesis planning to break down complex molecules into simpler starting materials.
- ☐ Synthesis planning has been made trivial by modern deep learning methods.
- ☐ Synthesis planning involves only selecting the reactants.

Question 23: Retrosynthesis as a reinforcement-learning game

Multi-step retrosynthesis can be cast as a game solvable by reinforcement learning. Which of the following statements about this “retrosynthesis game” are correct?

- ☐ The actions in the retrosynthesis “game” involve selecting reactants for the reactions.
- ☐ The state in the retrosynthesis “game” refers to the set of available building blocks.
- ☐ The environment in the retrosynthesis “game” performs the reaction and returns new molecules.
- ☐ The reward in the retrosynthesis “game” is obtained when the product is successfully synthesized from the building blocks.

Medical Data & Imaging

Question 24: Tabular medical data: types and missing values

Suppose we are given a table of patient medical records with three columns: “smoking status” (one of three string values: non-smoker, former smoker, current smoker), “cholesterol level” (floating-point measurements in mg/dL), and “diabetes risk score” (integer values in $[0, 10]$ indicating the likelihood of developing diabetes). Which of the following statements are correct?

- ☐ Missing values in “cholesterol level” can be replaced by the column’s median value to handle skewness.
- ☐ Column “smoking status” is continuous, as it captures the various smoking habits of patients.
- ☐ “cholesterol level” is ordinal since cholesterol measurements can be arranged in increasing order.
- ☐ Missing values in “smoker status” can be replaced by the column’s mean value to handle skewness.

Question 25: Batch effects in medical data

Which statements about batch effects in medical data are correct?

- ☐ Ignoring batch effects can lead to models that distinguish between batches rather than the actual biological conditions.

- ☐ Batch effects can be eliminated by simply increasing the sample size.
- ☐ Batch effects can create artificial differences in data due to variations in experimental conditions.
- ☐ Batch effects are usually beneficial as they introduce necessary variability into the data.

Question 26: Domain shifts in medical and healthcare data

Medical and healthcare data are frequently subject to domain shifts. Using the lecture notation (y = disease status, e.g. “healthy”/“diseased”; x = patient features, e.g. age, blood parameters, heart rate), which of the following statements about domain shifts are correct?

- ☐ If a machine learning model has a high predictive quality on a randomly drawn validation set, this means it will automatically adjust to domain shifts and maintain predictive quality over time.
- ☐ The distribution of the patient features x with distribution $p(x)$ is never affected by domain shifts.
- ☐ During disease waves of infectious diseases (such as COVID-19), the prevalence of a disease and thus the probability of observing this phenotype $p(y)$ changes.
- ☐ Assume that the diagnostic process to determine the disease status y is changed, e.g. an antigen test is replaced by a PCR test. This represents a concept shift $p(y | x)$.

Question 27: Medical imaging modalities

Which statements about medical imaging modalities are correct?

- ☐ Computed Tomography (CT) scans use ultrasound images to generate images of the body.
- ☐ X-rays are used to differentiate between tissues of varying densities, such as bone and soft tissue.
- ☐ Magnetic Resonance Imaging (MRI) provides detailed images of internal structures by using strong magnetic fields and radio waves.
- ☐ Optical coherence tomography imaging uses multiple X-rays to create images of organs and tissues.

Question 28: Most common task in medical imaging

Compared with typical computer-vision tasks, which task is especially prominent in medical imaging?

- ☐ Image generation.
- ☐ Classification.
- ☐ Segmentation.
- ☐ Object detection.

Question 29: The U-Net architecture

Which of the following correctly describe the U-Net architecture?

- ☐ The U-Net was the first vision model where the encoder is based on self-attention instead of convolutions.
- ☐ The U-Net architecture uses a symmetrical encoder and decoder.
- ☐ The U-Net has been specifically developed for the object detection task.
- ☐ To aid the decoding process, feature maps from the encoder are concatenated with feature maps of the corresponding up-sampling level of the decoder.

Question 30: The “Clever Hans” effect

In the context of AI for medical imaging, what does the “Clever Hans” effect refer to?

- ☐ A bias occurring when enhancing the resolution of low-quality medical images.
- ☐ The situation where an AI model appears to perform well but is actually relying on irrelevant features or artifacts in the data.
- ☐ The phenomenon where an AI system incorrectly predicts outcomes based on biased training data.
- ☐ The phenomenon where data augmentation is used to improve the robustness of AI models in medical imaging.

Question 31: Explainable AI and interpretability in medical imaging

Which statements about explainable AI and interpretability in medical imaging are correct?

- ☐ Gradient integration utilizes the propagation of a relevance value back to the input layers, layer by layer, to estimate the relevance of each input feature.
- ☐ Explainable AI techniques, such as Grad-CAM, help visualize which parts of a medical image are most important for a model's prediction.
- ☐ Occlusion analysis involves removing patches of the training data images to improve the model's interpretability.
- ☐ Counterfactual explanations in medical imaging involve generating alternative images to show what changes would have been needed for a different diagnosis.

Question 32: AlphaFold 2

Which of the following statements about AlphaFold 2 are correct?

- ☐ AlphaFold 2 incorporates quantum mechanical calculations to improve the accuracy of its predictions.
- ☐ AlphaFold 2 incorporates multiple sequence alignments and homologous protein structures to improve the accuracy of its predictions.
- ☐ AlphaFold 2 is a deep learning model designed to predict the three-dimensional structure of proteins from their amino acid sequences.
- ☐ AlphaFold 2 uses a convolutional neural network architecture.

Biological Sequences

Question 33: Composition of biological sequences

Which statements about the composition of biological sequences are correct?

- ☐ Protein sequences are composed of chains of amino acids linked by peptide bonds.
- ☐ DNA sequences are composed of nucleic acids.
- ☐ Biological sequences do not follow any specific patterns or rules in their composition.
- ☐ RNA sequences are composed of lipids.

Question 34: Deep learning for biological sequences

Which of the following statements about deep learning methods for biological sequences are correct?

- ☐ Feature extraction from biological sequences can be used to identify motifs, which are recurring patterns or elements within the sequences.
- ☐ One-hot encoding is commonly used to prepare biological sequences as input for deep learning models.
- ☐ Deep learning techniques can be applied to tasks such as classification, segmentation, and generative modeling of biological sequences.
- ☐ 1D-Convolutions, Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and Transformers are all used to analyze biological sequences.

2026 exam — single-best-answer

The remaining sections are from the **2026** exam (25 June 2026). Each of these questions has **exactly one** correct option among the three listed. Wording has been paraphrased; the answer options are kept intact.

Molecules, Representations & Cheminformatics (2026)

Question 35: Molecular graphs vs. fingerprints

Which of the following statements accurately describes the relationship between molecular graphs and molecular fingerprints?

- ☐ A fingerprint can be derived from a molecular graph, but the exact graph generally cannot be reconstructed from the fingerprint.
- ☐ Fingerprints and molecular graphs necessarily contain the same structural information.
- ☐ A molecular graph can always be reconstructed exactly from a fingerprint.

Question 36: Converting between SMILES and fingerprints

Which statement about converting between SMILES strings and molecular fingerprints is correct?

- ☐ Both SMILES and molecular fingerprints can be converted into each other using cheminformatics tools.
- ☐ Only a SMILES can be converted into a molecular fingerprint using cheminformatics tools.
- ☐ Only a molecular fingerprint can be converted into a SMILES using cheminformatics tools.

Question 37: Tanimoto coefficient

How does the Tanimoto coefficient quantify the similarity of two molecular fingerprints?

- ☐ It divides the number of shared features by the total number of features present in either molecule.
- ☐ It aligns three-dimensional pharmacophore points and calculates their average spatial distance.
- ☐ It measures the overlap between the core scaffolds and assigns a score of one whenever the scaffolds match.

Question 38: Aspirin SMILES

Which of the following SMILES strings represents aspirin (acetylsalicylic acid)?

- ☐ O=C(C)Oc1ccccc1C(=O)O
- ☐ cc(=O)Oc1ccccc1c(=O)O
- ☐ O=C[C]Oc1ccccc1C[=O]O

Question 39: Bioactivity datasets

Which description of typical bioactivity datasets is the most accurate?

- ☐ They are generally large, balanced, and nearly free of measurement noise.
- ☐ They are often limited, imbalanced, and noisy, with many more inactive than active compounds.
- ☐ They are inexpensive to generate and can therefore cover most combinations of compounds and assays.

Question 40: The Design–Make–Test–Analyse cycle

Which statement correctly describes the Design–Make–Test–Analyse (DMTA) cycle in drug discovery?

- ☐ Compounds are tested before they are designed, and only active compounds are subsequently synthesized.
- ☐ Compounds are designed and synthesized once, after which testing replaces any need for further design.
- ☐ Compounds are designed, synthesized, experimentally tested, and the results are analysed to guide the next design cycle.

Virtual Screening, QSAR & Targets (2026)

Question 41: Structure-based vs. ligand-based drug design

Which statement correctly distinguishes structure-based drug design (SBDD) from ligand-based drug design (LBDD)?

- ☐ SBDD requires known active ligands, whereas LBDD requires the 3D structure of the target.
- ☐ SBDD and LBDD both require a high-resolution 3D structure of the target.
- ☐ SBDD uses the 3D structure of the target, whereas LBDD can use information from known ligands without requiring that target structure.

Question 42: Similarity search vs. model-based search

Which statement correctly distinguishes similarity search from model-based search?

- ☐ Similarity search is ligand-based, whereas model-based search necessarily requires the three-dimensional structure of the target binding site.
- ☐ Similarity search retrieves compounds similar to a query molecule, whereas model-based search uses a supervised model learned from labelled examples to predict activity.
- ☐ Similarity search learns a decision function from labelled active and inactive compounds, whereas model-based search ranks compounds by proximity to a query molecule.

Question 43: Early-recognition metrics (odd one out)

Which metric is *not* specifically designed to emphasise early recognition of active compounds near the top of a ranked virtual-screening list?

- ☐ BEDROC.
- ☐ AUC-ROC.
- ☐ Enrichment factor.

Question 44: Evaluating a virtual-screening model

A virtual-screening model ranks one million compounds, but only the top 1% can be tested experimentally. Which evaluation strategy is most appropriate for selecting the model?

- ☐ Select the model with the highest accuracy after applying a 0.5 probability threshold, because ranking quality above that threshold is irrelevant once compounds are classified.
- ☐ Select the model with the highest global AUC-ROC, because equal weighting of all ranking positions guarantees the best enrichment within the top 1%.
- ☐ Prioritize an early-recognition metric such as enrichment factor near the 1% cutoff or BEDROC, because performance at the top of the ranking determines experimental utility.

Question 45: Defensible generalization estimates in QSAR

Which workflow gives the most defensible estimate of generalization performance in a QSAR pipeline?

- ☐ Define the data split first, perform feature and hyperparameter selection using training data, and use the held-out test set only for final evaluation.
- ☐ Split the data first, compare model and hyperparameter choices on the test set, and report the best test-set result.
- ☐ Select features on the complete dataset, then split the reduced data and tune hyperparameters using only the training subset.

Question 46: Compound-series bias

What is compound-series bias, and why is it problematic in QSAR modelling?

- ☐ Many compounds are highly structurally related, so a random split can place close analogues in training and test sets and underestimate generalization error.
- ☐ Information from test compounds is used during feature or hyperparameter selection, so the reported performance is biased even when compounds are structurally unrelated.
- ☐ The active class is much smaller than the inactive class, so performance is dominated by the majority class even when chemical series are diverse.

Question 47: Drug synergy and model symmetry

Which statement correctly describes drug synergy together with an appropriate property of a synergy-prediction model?

- ☐ The combined effect exceeds the additive effect of the separate compounds, and the prediction should not change when the order of the compounds is swapped.
- ☐ The combined effect equals the additive expectation, and the prediction may change when the same pair is supplied in reverse order.
- ☐ The combined effect is smaller than the additive expectation, and the prediction should be invariant to the order of the compounds.

Question 48: Proteochemometric modelling

What distinguishes proteochemometric modelling from classical single-target QSAR?

- ☐ It jointly uses representations of molecules and protein targets.
- ☐ It uses molecular representations with a softmax output over candidate protein targets.
- ☐ It uses molecular representations with a separate prediction head for each protein target.

Question 49: Target deconvolution / target fishing

What is target deconvolution, also known as target fishing?

- ☐ Given a protein and a molecular library, rank all molecules by predicted affinity to that one protein.
- ☐ Given a molecule and multiple candidate proteins, predict which protein the molecule is

most likely to bind.

- ☐ Given a molecule, independently predict binding to every candidate protein so that any number of targets may be assigned simultaneously.

Graph Neural Networks (2026)

Question 50: Reordering atoms in a molecular graph

The atoms of a molecular graph are relabelled (reordered) without altering its connectivity. How should a graph neural network behave?

- ☐ Node representations should be reordered in the same way, but the graph-level prediction may change because the adjacency matrix has been reordered.
- ☐ Both node representations and the graph-level prediction should remain in exactly the same tensor positions.
- ☐ Node representations should be reordered in the same way, while a graph-level prediction should remain unchanged.

Question 51: Steps of an MPNN layer

Which option correctly lists the typical steps performed in one Message Passing Neural Network (MPNN) layer?

- ☐ Message generation, aggregation, update.
- ☐ Message generation, graph convolution, aggregation.
- ☐ Graph convolution, message generation, update.

Question 52: Oversmoothing in GNNs

After stacking many message-passing layers, the node representations in a molecular graph become almost indistinguishable. Why is this problematic?

- ☐ Long-range information is compressed through narrow graph bottlenecks, so distant signals cannot be transmitted effectively.
- ☐ The model may lose the ability to distinguish local atom environments because node embeddings become too similar.
- ☐ Important information fails to reach distant nodes because too few message-passing steps were used.

Question 53: Contrastive learning

How does contrastive learning construct a shared molecule–target representation space?

- ☐ It clusters molecules and targets separately and assumes clusters with the same index correspond to binding pairs.
- ☐ It increases similarity for associated molecule-target pairs and decreases similarity for mismatched pairs.

- ☐ It concatenates molecule and target representations and fits a supervised activity regressor without comparing pairs in a shared space.

Question 54: Multiple Instance Learning

In what way does Multiple Instance Learning differ from standard supervised learning? Instead of feature vectors each labelled with a single value $(\mathbf{x}, y)$, ...

- ☐ ... we have a bag of feature vectors labelled with a single value $(\{\mathbf{x}_1, \dots, \mathbf{x}_N\}, y)$.
- ☐ ... we have a bag of feature vectors labelled with a bag of values $(\{\mathbf{x}_1, \dots, \mathbf{x}_N\}, \{y_1, \dots, y_M\})$.
- ☐ ... we have feature vectors labelled with a bag of values $(\mathbf{x}, \{y_1, \dots, y_M\})$.

Generative Models & Synthesis Planning (2026)

Question 55: Reward hacking a QSAR proxy

A generator is optimized against a QSAR-based activity score. It produces molecules with very high predicted activity, but follow-up assays show poor activity. What is the most likely explanation?

- ☐ The generator received the activity score only after completing each molecule, making the reward delayed.
- ☐ The generator exploited weaknesses of the proxy scoring model rather than improving true experimental activity.
- ☐ The generator produced too few structurally distinct molecules, reducing diversity among candidates.

Question 56: Combating analogue rediscovery (odd one out)

A goal-directed generator repeatedly rediscovers close analogues of earlier high-scoring molecules. Which intervention is *not* an approach to combat this behaviour?

- ☐ Mix samples from a fixed exploratory generator with samples from the trainable generator during SMILES generation.
- ☐ Penalize the reward for molecules that are similar to previously generated high-scoring molecules stored in a memory buffer.
- ☐ Increase exploitation by always selecting the molecule with the highest predicted score as the next training example.

Question 57: Autoregressive molecular reinforcement learning

In autoregressive molecular generation framed as reinforcement learning, which mapping of RL concepts is correct?

- ☐ The partial molecule is the state and the next construction step is the action, but the scoring function is the policy and the generator supplies the reward.
- ☐ The completed molecule is the state, the scoring function is the action, and each partial molecule receives an independent final reward.

- ☐ The partial molecule is the state, the next construction step is the action, the generator is the policy, and the completed molecule receives a reward.

Question 58: Diffusion vs. one-shot latent decoding

What distinguishes diffusion-based molecular generation from one-shot latent decoding?

- ☐ It maps a single latent vector directly to a complete molecule in one decoder pass.
- ☐ It starts from noise and iteratively applies a learned denoising process.
- ☐ It generates a molecule step by step by predicting the next token or graph action from the partial molecule.

Question 59: Conditional generation

A generator receives a desired logP value as an input and produces molecules in one pass. It was trained on molecule–logP pairs and does not evaluate a logP scoring function during generation. Which paradigm does this describe?

- ☐ Conditional generation.
- ☐ Goal-directed generation.
- ☐ Distribution learning followed by unconditional sampling.

Question 60: Distribution learning

Which statement best describes distribution learning in molecular generation?

- ☐ Distribution learning generates molecules conditioned on auxiliary information such as protein targets or desired property values.
- ☐ Distribution learning aims to generate molecules that resemble those in the training dataset by learning their underlying distribution.
- ☐ Distribution learning focuses on generating molecules that are optimized for specific biological or physicochemical properties.

Question 61: Mode collapse

Which statement best describes mode collapse in generative molecule models?

- ☐ The model generates molecules with high validity and novelty but poor agreement with the training-set property distribution.
- ☐ The model repeatedly generates the same or very similar molecules.
- ☐ The model generates almost exclusively molecules that already occur in the training dataset.

Question 62: Reaction prediction and retrosynthesis

Which mapping between task and input/output is correct?

- ☐ Both forward reaction prediction and single-step retrosynthesis take only reaction conditions as input and predict reaction yield.

- ☐ Forward reaction prediction maps reactants to products, while single-step retrosynthesis maps a target product to possible precursors.
- ☐ Forward reaction prediction maps products to reactants, while single-step retrosynthesis predicts the major product from known reactants.

Question 63: Reaction templates

What is the purpose of a reaction template?

- ☐ To provide a generalized symbolic rule describing required substructures and how reactants are transformed into products.
- ☐ To encode reaction conditions such as solvent and temperature, while leaving the structural transformation to a neural model.
- ☐ To assign a reaction class label that groups similar reactions, without specifying how atoms or bonds are transformed.

Question 64: Easiest reaction task

Which of these tasks is generally the easiest?

- ☐ Single-step retrosynthesis.
- ☐ Multi-step retrosynthesis planning.
- ☐ Forward reaction prediction.

Question 65: Template-free translation methods

How do template-free translation methods frame reaction prediction or retrosynthesis?

- ☐ As retrieval of the most compatible stored reaction pattern from associative memory.
- ☐ As classification of the input into a learned reaction category followed by symbolic transformation.
- ☐ As direct sequence-to-sequence translation between molecular reaction strings.

Question 66: Deep networks inside MCTS for retrosynthesis

Why are deep neural networks useful inside Monte Carlo Tree Search for retrosynthesis?

- ☐ They replace tree search by evaluating only the first retrosynthetic disconnection and then executing that route greedily.
- ☐ They can estimate the value of unvisited states and suggest promising actions when exhaustive search is infeasible.
- ☐ They mainly balance chemical equations at each node, while the search itself decides value and policy without learned guidance.

Question 67: Pocket-conditioned 3D ligand generation

Which statement best describes pocket-conditioned 3D ligand generation?

- ☐ Ligands are generated without protein context and docked into the pocket only after

generation; the docking score does not condition the generator.

- ☐ The protein binding pocket provides geometric context for generating or denoising a ligand, often followed by validity, docking, property, and synthesis checks.
- ☐ A fixed ligand graph is provided and only alternative conformations of that same molecule are sampled without using protein information.

Answer Key

Note on scoring

Questions 1–34 (the 2024 exam) are multiple-select: each listed letter is a correct option, and any option *not* listed is incorrect. On the original Moodle exam, each correct tick added points and each incorrect tick subtracted them. Questions 35–67 (the 2026 exam) are single-best-answer: exactly one letter is correct.

- Q1. A, B** — SMILES is the *Simplified Molecular Input Line Entry System* and is a linear character sequence. It *can* be converted to a graph, and is *not* unique (one molecule has many valid SMILES).
- Q2. A, B** — both O=C(C(=O)O)C and CC(=O)C(=O)O encode CH3-CO-COOH. CCC000 and OCC00 do not.
- Q3. B, C** — fingerprints are binary/numerical descriptors used for similarity search and ML. They are lossy (no unique graph reconstruction) and carry no 3D information.
- Q4. A, B, C** — bioassays measure a biological response, test at various concentrations (dose-response), and are wet-lab procedures. “*in vitro* = in a living organism” is false (*in vivo* is the living organism).
- Q5. A, C, D** — Lipinski’s Ro5 is an ADME proxy, clearance is hard to model, and ADME is central to pharmacokinetics. Good ADME does *not* guarantee an effective drug.
- Q6. C, D** — QSAR data often comes in structurally related series and is noisy. It is usually *small* (not thousands/millions) and *imbalanced* (not balanced).
- Q7. B, C** — QSAR predicts drug candidates and toxicity. 3D protein modeling and genome mapping are different problems.
- Q8. A** — fingerprints are standard in ligand-based VS. It *is* suitable for unknown compounds, does *not* need the protein structure, and considers more than bioactivity (ADME, filters).
- Q9. A, D** — temporal splits simulate real-world acquisition; cluster splits keep structurally similar compounds out of both sets. Random splits do *not* eliminate bias, and CV *is* used.
- Q10. B, D** — BEDROC and the Enrichment Factor reward early recognition. Precision-Recall curve and Average Precision are general metrics, not early-recognition-specific.
- Q11. A, B** — target identification finds the *most likely* target via multi-class classification. “All proteins” and “multi-task” describe target *prediction*, not identification.
- Q12. A, B** — synergy is “greater than the sum”; the Loewe additivity model assesses it. Antagonism is *less* than the sum, and synergy applies to any drug pair (not only similar ones).
- Q13. B, C** — RF overfits less and is more interpretable than GNNs. GNNs are *not* universally better and are *not* less sensitive to data quality.
- Q14. A, B** — a molecular graph is a node-feature matrix plus an edge/adjacency matrix; node features include atomic number, hybridization, charge. GNNs *do* use edge information, and bond type *can* be encoded.
- Q15. B, C, D** — GNNs struggle with long-range interactions, very large graphs, and are compute-heavy. They are *not* “incapable of predicting any property accurately”.

- Q16. A, B** — multi-modal contrastive learning maps modalities into a shared latent space and is *self-supervised*. It does not “concentrate representations to make predictions”, and is *not* unsupervised.
- Q17. C, D** — a MIL bag has a *single* label and the model must be permutation-invariant. Instances are *not* individually labelled, and bag size need *not* be fixed.
- Q18. D** — autoregressive models can generate SMILES. Modern generators produce mostly valid SMILES, generative models explore *novel* space, and GANs *can* generate molecules.
- Q19. C, D** — goal-directed generation targets *specific* desired properties and can be optimised with RL. Replicating existing structures / matching the training distribution describes *distribution learning*.
- Q20. B, D** — FCD compares means and covariances of *latent* (ChemNet) representations and measures distance between *distributions* of generated vs. real molecules — not individual atoms, not raw structures.
- Q21. A, B, C** — reaction templates, training on known-reaction databases, and transformer architectures are all used. QM data is *not* required.
- Q22. A, B** — synthesis planning designs a reaction sequence and uses retrosynthetic analysis. It is *not* trivial, and involves more than just picking reactants.
- Q23. A, C, D** — per the lecture’s framing the *action* is selecting reactants, the *environment* applies the reaction and returns new molecules, and the *reward* comes when the current set are all available building blocks. Only the *state* option is wrong: the state is the *current set of molecules* to be made, not the set of available building blocks.
- Q24. A** — cholesterol is numeric, so the *median* handles skewed missing values. Smoking status is categorical (not continuous/mean-imputable), and cholesterol is continuous, not merely ordinal.
- Q25. A, C** — ignored batch effects make models separate batches instead of biology, and they create artificial differences. They are *not* removed by more samples and are *not* beneficial.
- Q26. C, D** — changing prevalence is a label/prior shift $p(y)$; changing the diagnostic procedure is a concept shift $p(y | x)$. Good validation accuracy does *not* auto-adapt, and $p(x)$ *can* shift.
- Q27. B, C** — X-rays distinguish tissue densities; MRI uses magnetic fields and radio waves. CT uses X-rays (not ultrasound), and OCT uses light (not X-rays).
- Q28. B, C** — classification and segmentation are the dominant medical-imaging tasks (segmentation especially, vs. typical vision tasks).
- Q29. B, D** — U-Net has a symmetric encoder/decoder with skip connections (encoder feature maps concatenated into the decoder). It is convolutional (not attention-based) and was built for *segmentation*, not detection.
- Q30. B** — the Clever Hans effect is a model that *looks* good but relies on irrelevant features/artifacts (shortcut learning).
- Q31. B, D** — Grad-CAM highlights important image regions; counterfactuals show what change would flip the diagnosis. The first option mislabels LRP as “gradient integration”, and occlusion perturbs *input*, not training data.
- Q32. B, C** — AlphaFold 2 uses multiple sequence alignments + homologous structures to predict 3D structure from sequence. It uses an attention-based (Evoformer) architecture, *not* a CNN, and no QM calculations.

- Q33. A, B** — proteins = amino acids via peptide bonds; DNA = nucleic acids. Sequences *do* follow rules, and RNA is nucleic acid, not lipid.
- Q34. A, B, C, D** — all true: motif/feature extraction, one-hot input encoding, classification/segmentation/generation tasks, and 1D-CNN / RNN / LSTM / Transformer architectures.
- Q35. A** — a fingerprint is computed from the graph but is lossy/hashed, so the graph cannot be recovered; “same information” (B) and “exact reconstruction” (C) are false.
- Q36. B** — SMILES $\rightarrow$ fingerprint is standard; fingerprints are lossy, so fingerprint $\rightarrow$ SMILES is impossible.
- Q37. A** — Tanimoto = shared bits divided by the union of bits (intersection over union). B is 3D pharmacophore alignment; C is scaffold matching.
- Q38. A** — O=C(C)Oc1ccccc1C(=O)O is valid acetylsalicylic acid; B misuses lowercase (aromatic) atoms for the carbonyls; C misuses square brackets.
- Q39. B** — bioactivity data is limited, imbalanced, and noisy, with far more inactives than actives.
- Q40. C** — design, make (synthesize), test, analyse, then iterate to the next design.
- Q41. C** — SBDD uses the target’s 3D structure; LBDD uses known ligands without it.
- Q42. B** — similarity search retrieves neighbours of a query; model-based search uses a supervised model learned from labels.
- Q43. B** — AUC-ROC weights the whole ranking equally; BEDROC and the enrichment factor are early-recognition metrics.
- Q44. C** — use an early-recognition metric (enrichment factor near the 1% cutoff, or BEDROC).
- Q45. A** — split first, select features/hyperparameters on training data only, then evaluate on the held-out test set exactly once.
- Q46. A** — structurally related analogues split across train and test under a random split, giving an over-optimistic estimate.
- Q47. A** — synergy is super-additive, and a synergy-prediction model should be order-invariant (symmetric in the two compounds).
- Q48. A** — it jointly represents molecules *and* protein targets (single-target QSAR uses the molecule alone).
- Q49. B** — given a molecule and several candidate proteins, predict the most likely target.
- Q50. C** — node representations permute with the relabelling (equivariance), while the graph-level prediction is unchanged (invariance).
- Q51. A** — message generation, aggregation, update.
- Q52. B** — oversmoothing: node embeddings become too similar, so local atom-environment information is lost (A is over-squashing; C is too few layers).
- Q53. B** — pull associated molecule–target pairs together and push mismatched pairs apart.
- Q54. A** — a bag of feature vectors carrying a single shared label.
- Q55. B** — reward hacking: the generator exploited weaknesses of the proxy score rather than improving true activity.

- Q56. C** — always picking the single highest-scoring molecule increases exploitation and worsens rediscovery, so it is *not* a remedy (A and B both add diversity).
- Q57. C** — partial molecule = state, next step = action, generator = policy, completed molecule = reward.
- Q58. B** — diffusion starts from noise and iteratively denoises (A is one-shot decoding; C is autoregressive).
- Q59. A** — conditional generation: conditioned on the target property, one pass, no scoring function queried.
- Q60. B** — distribution learning models the training distribution and generates molecules resembling it.
- Q61. B** — mode collapse: the model repeatedly produces the same or very similar molecules.
- Q62. B** — forward prediction maps reactants $\rightarrow$ products; single-step retrosynthesis maps product $\rightarrow$ precursors.
- Q63. A** — a reaction template is a generalized symbolic rule giving required substructures and the reactant $\rightarrow$ product transformation.
- Q64. C** — forward reaction prediction is generally the easiest (single-step retro is one-to-many; multi-step planning is hardest).
- Q65. C** — template-free methods frame the task as direct sequence-to-sequence translation between reaction strings.
- Q66. B** — deep networks estimate the value of unvisited states and suggest promising actions when exhaustive search is infeasible.
- Q67. B** — the binding pocket provides geometric context to generate/denoise the ligand, usually followed by validity/docking/property/synthesis checks.